

Family name: _____ Given name: _____

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1. Consider a multidimensional schema with a **dimension that does not have any attribute** or aggregation hierarchy (e.g., order number). Briefly **justify** whether you would implement it like:
 - (a) A dimension table with only the primary key, pointed from the fact table.
 - (b) Only an attribute in the fact table, but not pointing to any dimensional table.
 - (c) No corresponding attribute or table in the star schema.
 - (d) Other option

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2. Consider an integrated system with two different sources (e.g., Vueling-V and Airbus-A). The tables are the following and the mapping between the local and global schemas is given using **GAV** with relational algebra expressions as follows (“get_date” is a function with the obvious meaning):

Vueling Hops(from, to, timestamp, plane, mainIncident)

Airbus Planes(registry, sits)

- $Flights(origin, destination, day, aircraft) := V.Hops[from, to, get_date(timestamp), plane]$
- $Incidents(aircraft, day, ATAcodes) := V.Hops[plane, get_date(timestamp), mainIncident]$
- $Aircrafts(aircraft, capacity) := A.Planes[registry, sits]$

Assuming that both databases reside in the same RDBMS (i.e., wrappers are not needed), give the **SQL query** generated by the mediator on receiving the following global one:

```
SELECT DISTINCT f.day, f.aircraft
FROM Aircrafts a
  JOIN Flights f ON a.aircraft = f.aircraft
  JOIN Incidents i ON f.aircraft = i.aircraft AND f.day=i.day
WHERE a.capacity > 200 AND i.ATAcodes=21 AND f.origin='BCN';
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3. Consider an RDBMS with a fact table **sales** with a boolean dimension **local** indicating whether the corresponding sale was “local” (true) or “global” (false). The table contains 10,000,000 rows of 80 bytes each in the average (including both data and metadata). The dimension has a **bitmap index** over its **local** attribute. Assuming a disk block of 8K,¹ give the **cost** of the following query and explicit any assumption you make:

```
SELECT SUM(income)
FROM sales
WHERE local;
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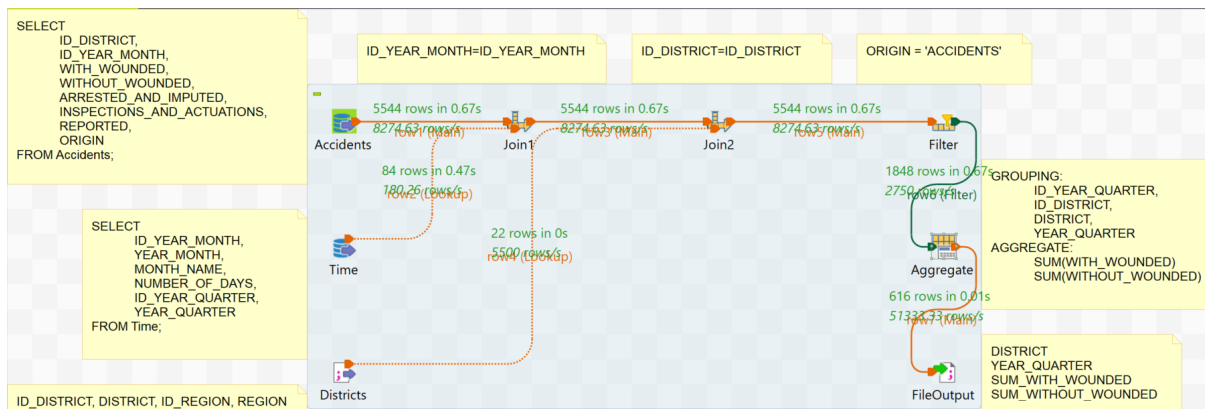
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4. Given the ETL below, where the stickers indicate the attributes and conditions of the nearest node, briefly explain how you would optimize its performance (a.k.a. execution time) or justify why you think it is already optimal. Explicit any assumption you need to make and annotate any change in the same diagram to facilitate the explanation.



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¹Consider 1K=1000bytes.

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5. Determine the liveness of the next three tables. Briefly justify your answer.

```
create table Teams (
name varchar(25),
budget int,
city varchar(15) not NULL,
primary key (name),
check (name='Futbol Club Barcelona ')
);
```

```
create table Players (
name varchar(30) not NULL,
age int check (age>0 and age<50),
team varchar(25) not NULL,
number int check (number>=0 and number<100),
primary key (name),
foreign key (team) references teams(name)
);
```

```
create table Matches (
local varchar(25),
visiting varchar(25),
primary key (local, visiting),
foreign key (local) references teams(name),
foreign key (visiting) references teams(name),
check (local<>visiting)
);
```

1. Teams (yes/no): _____

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2. Players (yes/no): _____

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3. Matches (yes/no): _____

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